

JORGE DOMINGUES NUNES

AI Engineer

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Pombal, Portugal.

My recent work and integrations have positioned me at the forefront of LLM engineering and agentic workflows. By building bridges between automation, I've enabled sophisticated AI agents to orchestrate complex business processes autonomously. This experience has given me deep expertise in building retrieval-augmented generation (RAG) pipelines that connect LLMs to live business data. Designing agent tooling and semantic layers that allow AI to reason over knowledge graphs and dynamic workflows, implementing vector search architectures with embeddings, retrieval strategies, and context management, creating cognitive memory systems where agents maintain state, learn from interactions, and make context-aware decisions.

EXPERIENCE



Founder & AI Solutions Architect
11/2025 - Present
www.api.iamjorgenes.com

Building production-ready agentic AI marketplace and orchestration platform. Full-stack ownership: Terraform IaC, Kubernetes pods, Python FastAPI backend, React frontend, Flutter mobile (iOS/Android). Multi-model support (Ollama, Anthropic, OpenAI-compatible), RAG pipelines, per-user memory, 32 deployment presets, and ISO/IEC 42001 compliance. Live at www.api.iamjorgenes.com.

- Infrastructure as Code & Kubernetes: Designed Terraform preset system with 32 deployment presets across 4 role categories (Data Provider, Token Consumer, Platform Operator, Agent Developer) supporting Docker, AWS, GCP, and Azure. Architected K8s cluster with custom CRDs for agent pods, Kustomize overlays for staging/production, Helm charts, and Ansible playbooks for post-deploy continuous configuration.
- Multi-Model AI & RAG: Integrated multiple LLM providers—Ollama (Mistral 7B, Llama 3.2, Qwen 2.5-Coder, DeepSeek-Coder) as primary with Anthropic Claude as fallback. Built WebRAG system with knowledge base management, hybrid search, per-user conversation memory (session-based 30-min persistence), token usage tracking per user, and intelligent caching achieving 90% cost reduction.
- Backend & Agentic Systems: Built Python FastAPI backend (200+ endpoints) with LangChain/LangGraph multi-agent orchestration, ChromaDB vector storage, Prometheus monitoring. Implemented 6 user roles (Data Provider, Token Consumer, Platform Operator, Agent Developer, Enterprise, Public) with tiered subscriptions (Free/Basic/Pro/Enterprise), token-based billing, and wallet management.
- Frontend & Mobile: Developed React web dashboard and Flutter cross-platform mobile application (iOS/Android) with embeddable chatbot widget (session persistence, rate limiting, domain allowlisting, theme customization). Integrated n8n for workflow automation and Langfuse/MLflow for LLM observability and prompt management.
- Compliance & Agent Marketplace: Implemented ISO/IEC 42001 AI governance with risk classification, CSA STAR/SOC 2 Type II audit logging, GDPR privacy controls, and ESG metrics tracking. Built agent marketplace with multiple templates (customer support, agentic assist) featuring LangGraph tool integrations (Terraform, Docker, MCP server, UI generation) and Ollama-powered local inference.



Principal AI Engineer
01/2024 - 10/2025
[Keyprog](http://www.keyprog.com)

Architected and deployed production-grade LLM-powered e-commerce platform with AI-driven content management. Built multi-agent systems for automated business workflows using enterprise-scale AI integration.

- AI Integration:** Developed MCP (Model Context Protocol) server with 29+ AI tools enabling natural language database interactions. Implemented RAG pipelines, prompt engineering for hallucination reduction, and LLM evaluation frameworks.
- Cloud Infrastructure: Designed scalable backend services using Directus, Docker, Kubernetes (AKS), PostgreSQL, Redis. Established LLMOps best practices including AI observability, automated evaluation with MLflow, and performance monitoring.
- Technical Leadership: Mentored engineering team on LLM integration patterns, conducted code reviews, led design discussions. Established CI/CD pipelines with GitHub Actions and security protocols for AI systems.
- Security & Access Control: Designed role-based permission system with granular

Senior AI Engineer with 7+ years of commercial software development and deep expertise in production-grade LLM systems beyond MVPs. I specialize in deploying and maintaining LLM-powered applications at scale in enterprise environments.

Deep proficiency in Python (FastAPI, Flask) with async programming. 3+ years building RAG pipelines, multi-agent architectures, and AI agents using OpenAI, Anthropic, Google Gemini, and LangChain/LlamaIndex. Strong LLMOps expertise: AI observability, hallucination mitigation, prompt engineering, MLflow evaluation, and performance monitoring. Skilled in Docker, Kubernetes (AKS), Helm, CI/CD pipelines, and cloud platforms (Azure, AWS, GCP). Experience with Apache Spark for large-scale data integration and model deployment pipelines.

Proven ability to mentor engineers and lead complex, distributed AI projects. Strong grasp of vector databases (Pinecone, Qdrant, ChromaDB), retrieval-augmented generation, hybrid search, and Chain-of-Thought/ReAct prompting methods. Experience with AI safety, governance, and compliance frameworks.

LANGUAGES

English	Fluent	●●●●●
French	Basic	●●●●●
Portuguese	Native	●●●●●
Spanish	Basic	●●●●●

SKILLS

CORE COMPETENCIES: LLM Production Systems, RAG & Vector DBs, Multi-Agent Systems, LLMOps & Observability, Technical Leadership

FRAMEWORKS: LangChain, LangGraph, Ollama, MLflow, TensorFlow, Huggingface, FastAPI

CLOUD: Azure (AKS), AWS, GCP, Databricks
VECTOR DBs: Pinecone, Qdrant, ChromaDB
DEVOPS: Docker, Kubernetes, Helm, CI/CD, GitHub Actions, Terraform, Ansible, Lamba
LLM APIs: OpenAI, Anthropic, Google Gemini
LANGUAGES: Python, TypeScript, Node.js, Dart/Flutter, SQL

Computer Vision (FastAPI, model training, MongoDB, Pydantic, Chunked Uploads, matplotlib, seaborn, Pytest, Pytest-asyncio)

[RAG-PDF \(Python, a Retrieval-Augmented Generation \(RAG\) project using LangGraph and LangChain\)](#)

[Mediapipe osc td](#) (Python, venv, websockets, numpy, opencv-python)



Senior AI/ML Engineer

Co2Offset.ai

04/2023 - 12/2024

Pombal, Portugal

- CO2OFFSET.AI is a SaaS solution that automatically measures the carbon captured in forests in a cheap, fast and reliable way, helping to grow and maintain forests by rewarding landowners for their protection.
- Cloud, IaC & Kubernetes: Terraform for multi-environment deployments, Ansible for configuration management. Kubernetes cluster orchestration with K8s Dashboard for pod monitoring, health checks, and resource scaling. Managed containerized microservices across staging/production pods with Helm charts, ConfigMaps, and Secrets. 200+ containerized Python endpoints (FastAPI, Docker). AWS architecture (IAM, EC2, S3, RDS, SES). CI/CD in GitHub Actions.
- AI/ML & Frontend: Integrated computer vision models and LLM frameworks deployed



Technical Consultant

European Central Bank

03/2020 - 03/2023

Frankfurt am Main, Germany

- Contributed to the latest main EBC website redesign and maintenance of the core Content Management System. Established robust privacy and security measures and ensuring data consistency across the platform.
- Partnered closely with the multimedia team to deliver seamless real-time integration of the President's live streaming, enhancing staff engagement and media coverage.
- Drove a large-scale website redesign powered by a Java code handled millions HTML files of articles published in multilingual setups and internationalization i18n modules, working directly with the ECB's Design team and a tech stack of JavaScript, HTML, Twig, Webpack, NPM, Grunt, SCSS, Angular, PHP, and MySQL, Microsoft Active Directory, Powershell scripting across Microsoft's Active Directory and multiple servers of the organization infrastructure, MVC and SOLID coding methodology
- Streamlined version control and releases via GitLab, JIRA, ensuring smooth pull request workflows deployment cycles.
- Implemented comprehensive end-to-end testing strategies for modules and services, boosting reliability and reducing regressions
- Configured and maintained Red Hat Linux servers, Apache virtual hosts for development team and deployment stages SSL SHA- 356 certificates, resulting in robust, secure development environments, Bash and Python scripting.
- Reinforced security protocols by applying Content Security Policy (CSP), mitigating XSS and other securing servers to prevent unauthorized access.
- Implemented Elasticsearch multilingual search engine capable of indexing and crawling server content in 24 languages, serving millions of pages, emails and links in real-time to support diverse user needs both internal CMS and public Website.
- Produced thorough technical documentation, ensuring seamless knowledge transfer and minimizing gaps between cross-functional teams

Senior Drupal Developer

European Commission

04/2018 - 01/2020

Brussels, Luxembourg

- Contributed to the main website redesign Project and oversaw Drupal migrations
- Designed a relational DBMS for tens of thousands of users.
- Bridged technical and business needs.
- Coordinated with project managers, developers, and stakeholders.
- Develop and optimize MySQL queries and Apache configurations.
- Developed batch processing API tools and Drupal, Drush-cli and Modules.
- Reduced deployment times and configuration errors.
- Implemented unit tests with Jest and React Testing Library.
- Integrated CI/CD pipelines.
- Automated builds and tests with Git and Jenkins.
- Maintained consistent environments.
- Followed accessibility standards.
- Collaborated to reduce downtime.
- Integrated dynamic content to boost engagement.
- Applied Agile methodologies with Confluence.
- Recognized for analytical thinking and proactive troubleshooting.
- Communicated clearly with all audiences. SEO
- Follow WCAG &ARIA guidelines to ensure Drupal accessibility compliance

- ✓ CSA STAR (Cloud Security Alliance)
- ✓ SOC 2 Type II (Service Organization Control)
- ✓ ISO/IEC 30134 (ESG Metrics for Data Centers)
- ✓ ISO/IEC 27001 (Information Security Management)
- ✓ GDPR (Data Privacy & Protection)

EDUCATION



Instituto Politécnico de Viana do Castelo

- Advanced coursework in Mathematics, Linear Algebra, and Calculus.
Extensive programming experience in C, C++, and Java, focusing on object-oriented design and real-time systems.
Expertise in 3D graphics programming, OpenGL, Direct X hardware accelerated real-time GPU computer graphics, AutoCAD, UML. UX/UI design principles
- Information Systems Programming
- IT Security, Technical English, Technical English, Communication Networks.
- Programming and Algorithms (introduction to programming logic)

CERTIFICATIONS & CONTINUOUS LEARNING

- Kubernetes Administration – Container orchestration, Helm charts, pod lifecycle management, ConfigMaps & Secrets.
- AI/ML Engineering – LLM fine-tuning, RAG pipelines, prompt engineering, LangChain, vector databases (Qdrant, Pinecone).
- Terraform & Infrastructure as Code – Multi-environment provisioning, Ansible automation, cloud-native deployments.
- Flutter & Cross-Platform Development – Mobile-first architecture, Dart, state management, REST API integration.
- CI/CD Pipelines – GitHub Actions, Jenkins, Docker multi-stage builds, automated testing and deployment workflows.

KEY PROJECTS

- **Exponential AI Platform:** Full-stack AI agent platform with Python backend, React frontend, and Flutter mobile app. Integrated LangChain for RAG pipelines, Ollama for local model inference, per-user memory and knowledge base management. Deployed on Kubernetes with auto-scaling pods and health probes.
- **E-Commerce Platform (Romance in Room):** Dropshipping e-commerce solution built with Directus headless CMS backend, Flutter storefront, and Odoo ERP integration for automated invoicing. Multi-environment deployment (staging/production) on Hetzner VPS with Docker containerization.
- **CI/CD MCP Pipeline:** Multi-Environment MCP Server enabling seamless bridge between localhost development and production VPS. Automated deployment pipelines with Docker, GitHub Actions, and infrastructure-as-code provisioning across staging and production environments.
- **Keyprog SaaS Platform:** Directus-powered headless CMS with custom collections, relationship mapping, automation flows, and role-based access control. Built RESTful APIs for data management, schema analysis, and bulk operations.

TECHNICAL HIGHLIGHTS

AI & Machine Learning: Production LLM systems with multi-model support (OpenAI, Anthropic, Google Gemini, Ollama). RAG pipelines with ChromaDB/Qdrant vector stores, LangChain/LangGraph agent orchestration, knowledge base management, per-user conversation memory, and hallucination mitigation strategies. MLflow for experiment tracking and model evaluation.

Cloud & Infrastructure: Kubernetes cluster management with K8s Dashboard, Helm charts, and pod health monitoring. Terraform multi-environment IaC, Docker containerization, AWS/GCP cloud platforms. CI/CD pipelines with GitHub Actions, Jenkins, and automated testing workflows. Hetzner VPS administration with Ansible configuration management.

Full-Stack Development: Python (FastAPI, Flask), TypeScript/React, Flutter/Dart cross-platform mobile. Directus headless CMS, PostgreSQL, MongoDB, Redis. MCP (Model Context Protocol) server development for AI tool integrations. RESTful API design with OpenAPI/Swagger documentation.